Rockborne

Crime Report

EDA of crime data in 4 major metropolitan areas

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Introduction

This document is to covers an overview of some exploratory data analysis (EDA) on a series of crime datasets across the UK assigned by Nadine Green. The aim of the project is to provide an analysis of crime within 4 major metropolitan areas and their corresponding police forces:

- Greater London - Metropolitan Police Service
- Merseyside – Merseyside Police
- West Midlands – West Midlands Police
- Thames Valley – Thames Valley Police

These are 4 key areas were chosen due to their relative importance in terms of population density and thus the overall size of their corresponding police force. The result of this EDA should be to highlight 2 major police forces that might potential need further studying, potentially requiring further analysis to support a decision regarding whether a new location should be opened there.

Datasets studied

There were 2 main datasets that were used to support the main claims made through the rest of the EDA:

- Crime and outcome data from https://data.police.uk/data
    - All 4 Prior listed police services
    - Date ranges from 2024-01 to 2026-01
- Population data split by age and gender from https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/adhocs/3194populationestimatesforpoliceforceareasinenglandandwalesbysingleyearofageandsexmid1991tomid2024?utm_source=copilot.com
    - Only used 2024 population data for accuracy

These 2 were chosen for their size, both sufficiently containing on the order of 100,000s to prevent a small sample size, and data granularity, each individual crime was given in the case of the crime and outcome data as well as the population data covering all ages from 0-85 inclusive and split further by gender.

The reason only 2024 data was used for population was due to the difficulty in obtaining reliable data on more recent population data and in order to maintain a good balance between recency and accuracy.

While LSOA population data would have achieved a similar result by grouping all police that hold jurisdiction per LSOA, the data provided would end up being too granular

compared to the chosen police force data which would have resulted in requiring more pre-processing than was necessary for the EDA to progress.

Data Dictionaries

Crime and outcome data

Table Column	Data Type	Description
Crime ID	String	A unique identifier for each crime, some minor crimes do not have a crime ID.
Month	Date	Month in which this crime occurred stored in the format of YYYY-MM.
Reported By	String	Police force that originally reported the crime.
Falls Within	String	Police force who have jurisdiction where the crime took place.
Longitude	Float	Longitude coordinate where the crime took place.
Latitude	Float	Latitude coordinate where the crime took place.
LSOA code	String	Lower layer Super Output Area (LSOA) code. Defines smaller areas for further population analysis
LSOA name	String	Name corresponding to the LSOA code.
Last Outcome type (Crime) / outcome type (outcome)	String	Action taken if any action was taken.
Context (crime only)	String	Any additional context surrounding the incident.

Population data

Table Column	Data Type	Description
Force	String	Police force of concern
Year	Integer	Year when this population data was collected
Ages… (Multicolumn given by GenderAge format e.g. F20)	Integer	Number of people at each age and gender

Preprocessing

While the preprocessing for both main datasets was relatively similar, there were some small differences due to the way that the initial data was structured.

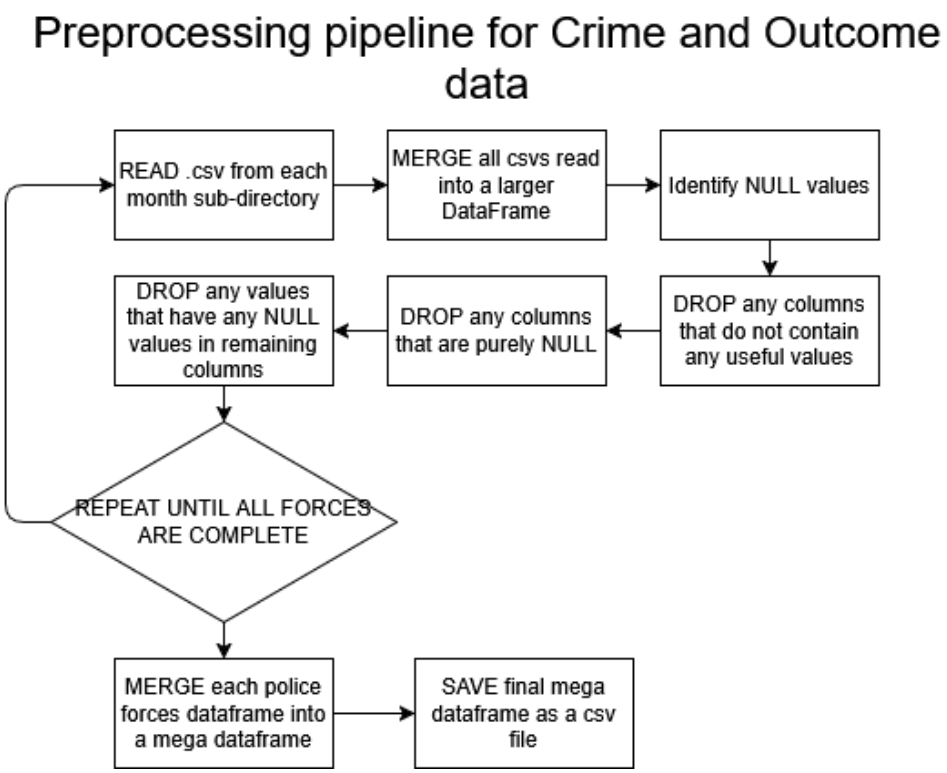
The end goal for the preprocessing phase was to generate a new set of CSVs that contained clean and ready-to-use data for the purposes of exploration through graphs and KPIs to further the study.

Another key tenet of the preprocessing was to leave data changes minimal, while grouping data for purposes of describing generalised trends would be useful, by leaving the data in its rawest state means that there would not be any potential data oversights be abstracting away from the original raw data.

Crime and outcome dataset

The main preprocessing pipeline was relatively similar across both the crime and the outcome datasets. The main task was to successfully read all the data that had been split into subdirectories and then further split into CSVs by police force, the data first needed to be merged before any cleaning and preprocessing was required.

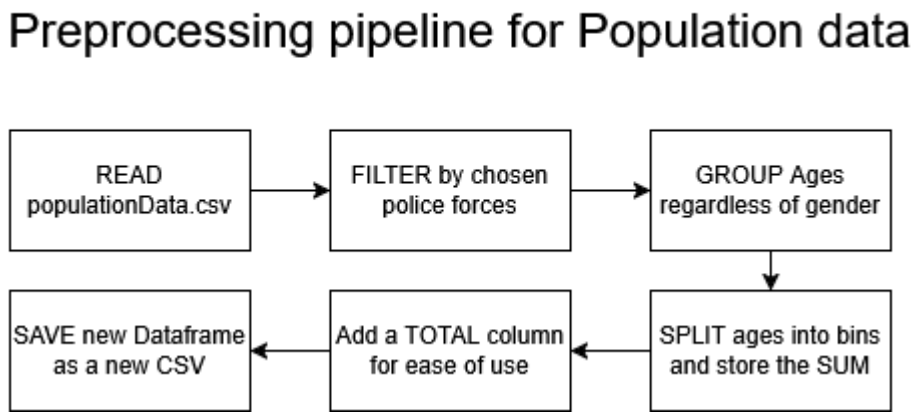
Preprocessing flow:



Population dataset

In comparison to the crime and outcomes dataset, the population dataset was contained within a single .xlsx file that needed to be extracted out into a csv file. Then, further processing was needed to merge the columns between male and female into a single column per age. Finally, each age was grouped into an arbitrary bin for slightly better abstraction of the data as full precision was not necessary for proper analysis.

Preprocessing flow:



Overall changes made after preprocessing

Change Made	Justification
Removed “Context” column from crime dataset.	The “context” column was all NULL values
Removed “Location” column from crime and outcome dataset.	The “Location” column is superseded by the Latitude and Longitude columns providing more accurate information
Removed all crimes with NULL latitude and longitude from the crime and outcome dataset.	Crimes that had missing latitude and longitude values made up a marginal amount of the datasets and these columns are necessary for further analysis.
Added totals column to population	The extra column was needed to prevent repeat calculations during the analysis phase.
Grouped population into different age bins.	There was no need for the granularity of each specific age and so the ages were binned to provide a better overview.

Analysis

The following shows the process that was undertaken to explore the data and provide some insights into further analysis:

NOTE: all numerical comparisons were done per 1000 people to ensure fairness regardless of population count.

Comparing each regions relative crime

The following shows a subset of crimes reported across each region, the darker the colour represents a greater density of crimes reported.

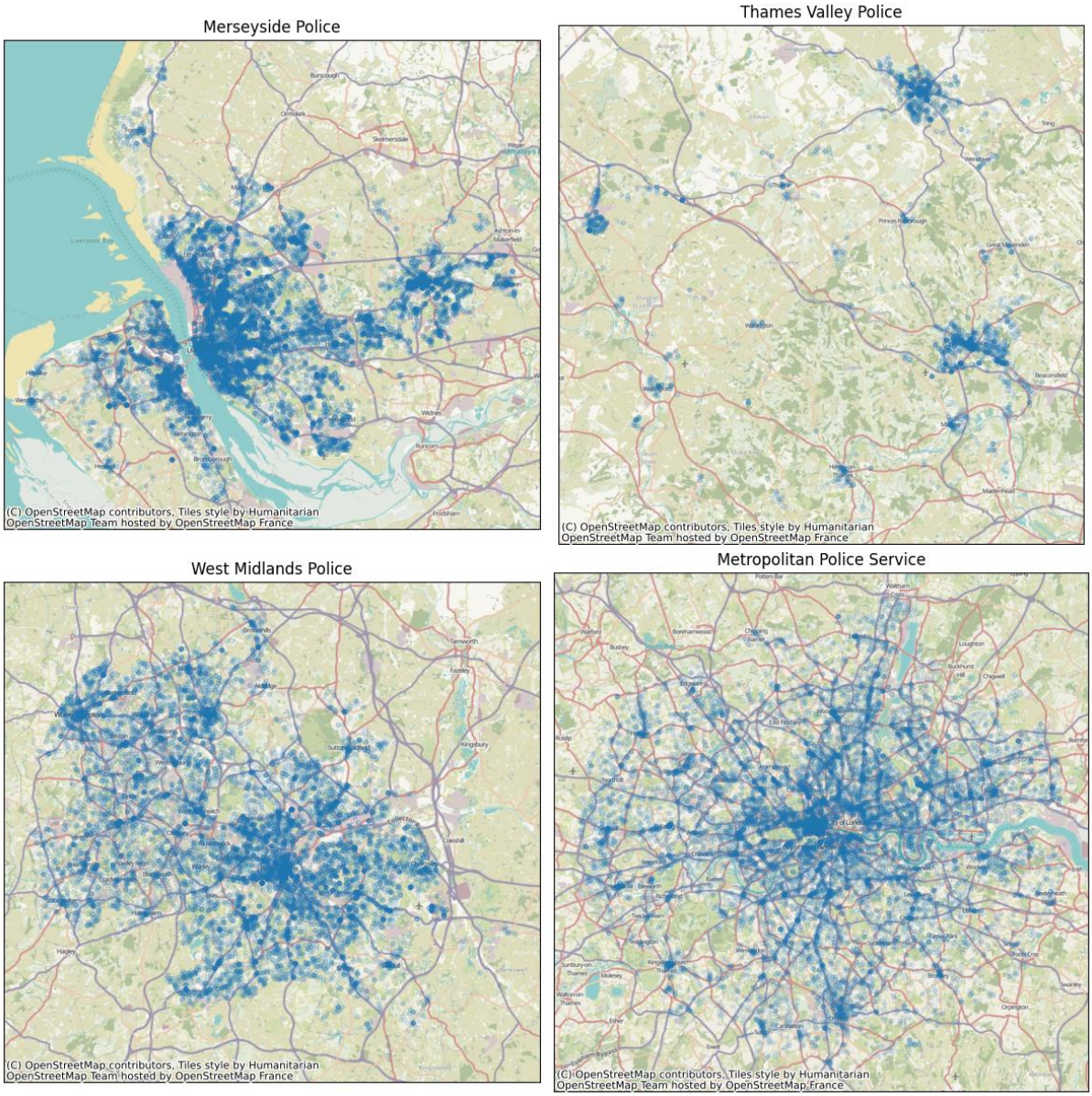


Figure 1: Crime across each region

From each plot, it is apparent that crimes recorded is proportional to the population density with a majority of the crimes recorded being within any denser region of

population. However this is not enough to compare each region, the following shows a plot where the size of the marker corresponds to the amount of crimes recorded per 1000 people. This metric was chosen due to the differing population sizes between the 4 major areas that were selected for study.

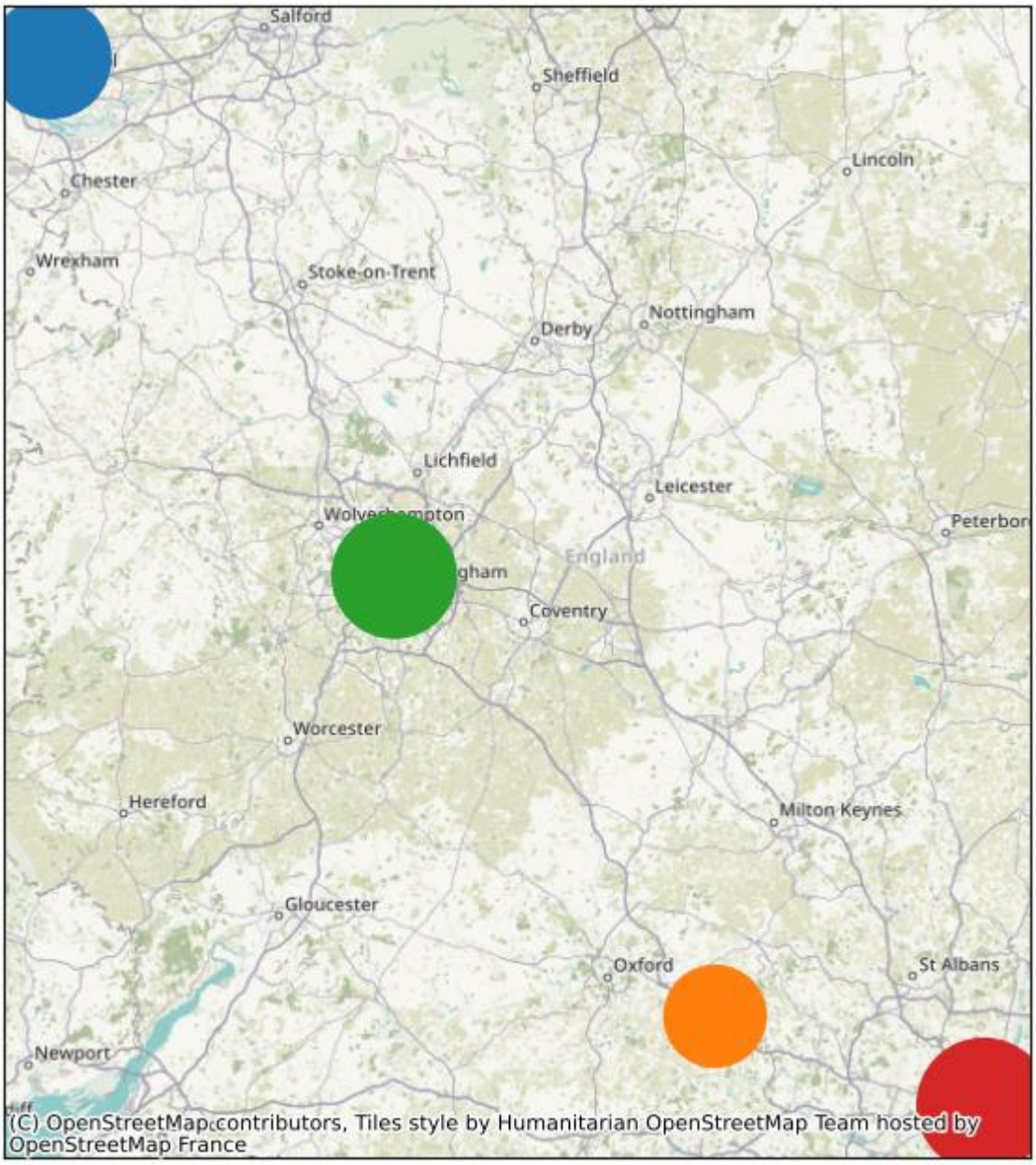


Figure 2: Comparing crime per 1000 people between Metropolitan (red), Thames Valley (Orange), West Midlands (Green) an Merseyside (blue)

Metropolitan	Thames Valley	West Midlands	Merseyside
260.7	151.3	226.8	213.7

As shown in Figure 2, the total crimes recorded in the past 2 years per 1000 population are relatively the same across all regions except for the Thames Valley region where the

crimes recorded are smaller, likely due to the lower population density of the countryside and smaller towns.

Changes in crime across time

The next exploration angle was to how crimes changed and their according outcomes in which action was taken.

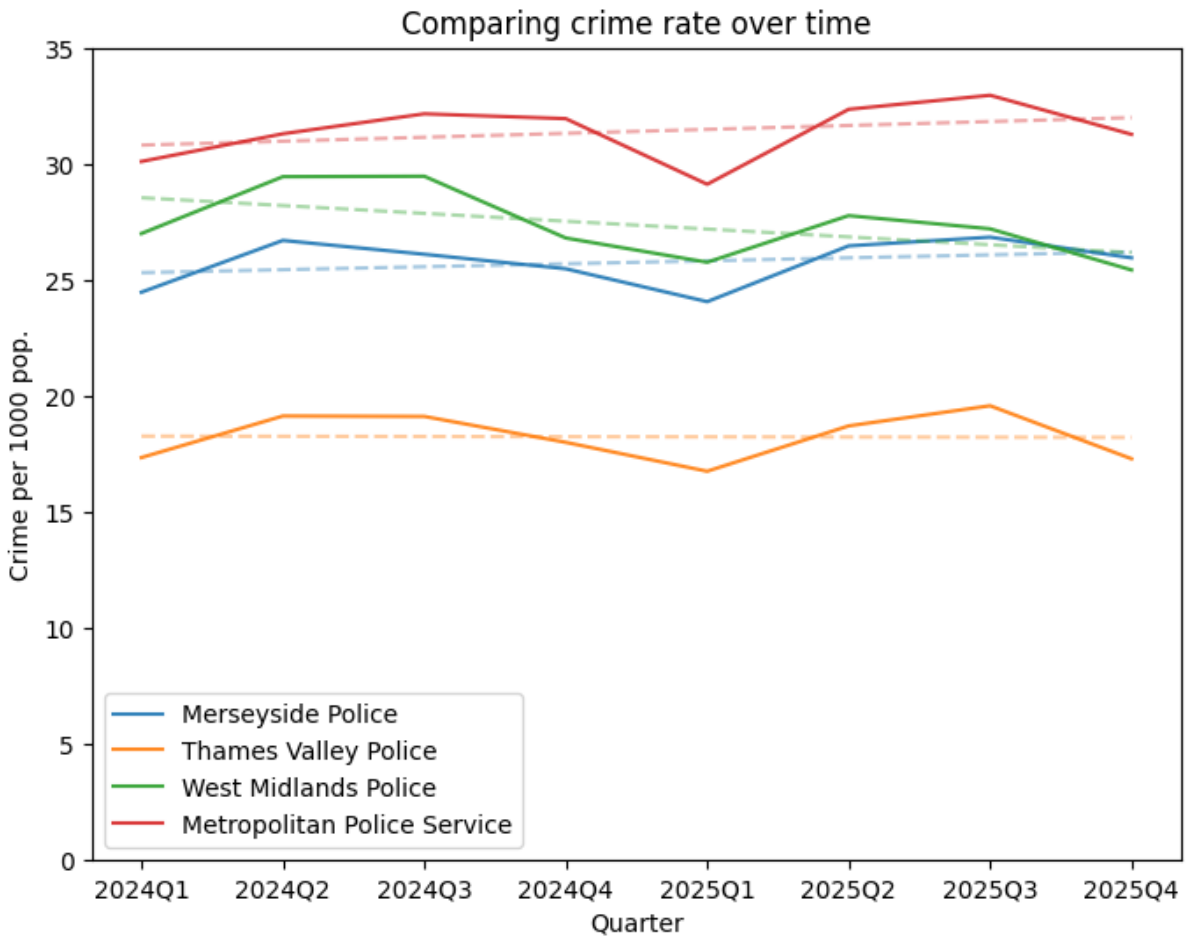


Figure 3: Comparing the growth of crime among all police forces

It is interesting to note that the 3 larger police services (Metropolitan, West Midlands, Merseyside) are all relatively similar values and all but the west midlands have a very slight increase in crimes recorded on a quarterly basis. The potential causes of the decline in recorded crimes in the west midlands might be needed for future studies.

However, it was not sufficient to only compare the reported crimes but also compare how many actions were taken in relation to each crime between the different services.

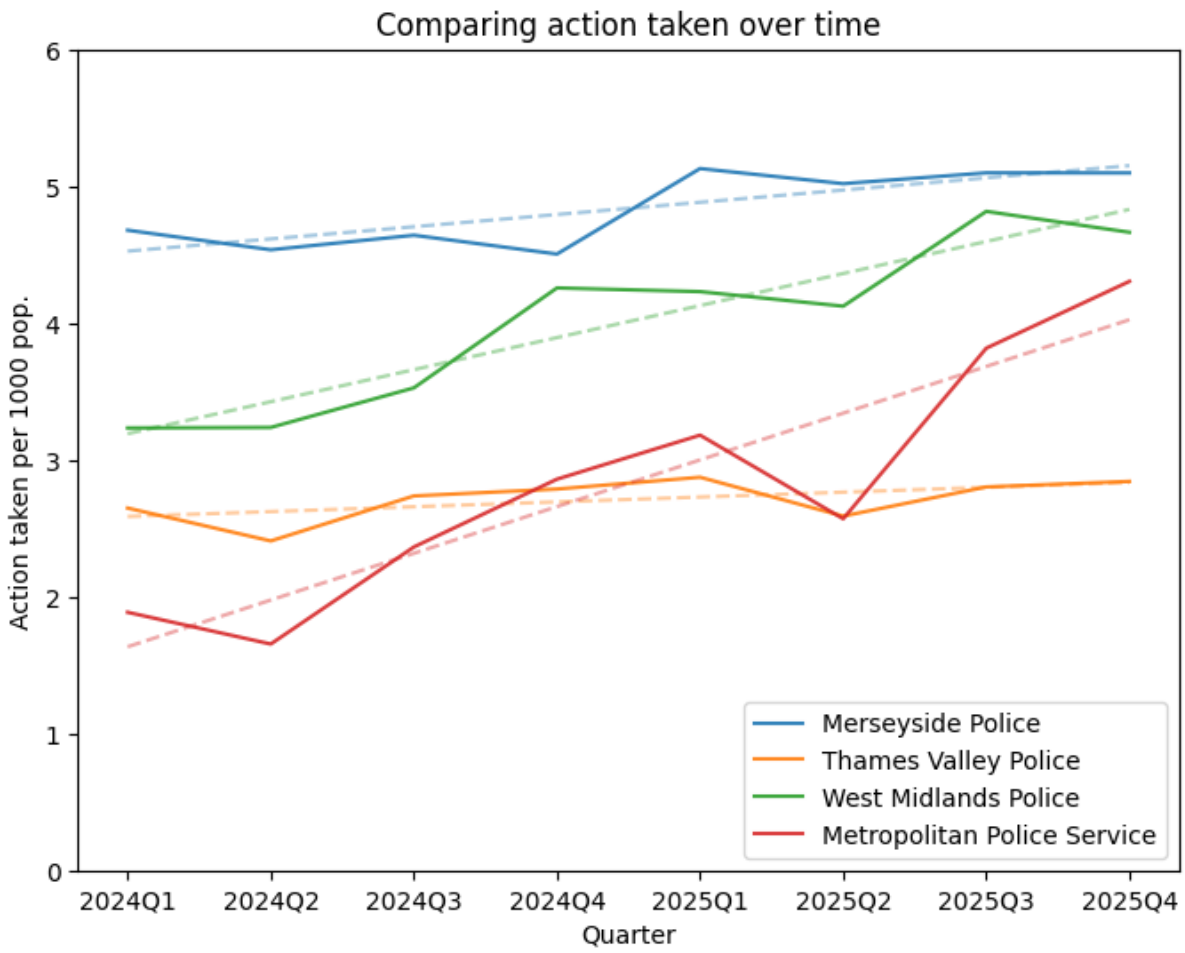


Figure 4: Comparing the growth of action taken outcomes across all police forces

Here, it is obvious while the Thames Valley police are maintaining relatively the same number of actions taken, the Metropolitan Police Service, West Midlands Police and the Merseyside police have been more aggressive with their actions taken. The former of which has rapidly improved with the sharpest rate of increase amongst all the police forces, this rapid improvement could be used as a basis for further study into the Metropolitan Police Service.

Comparing outcomes taken

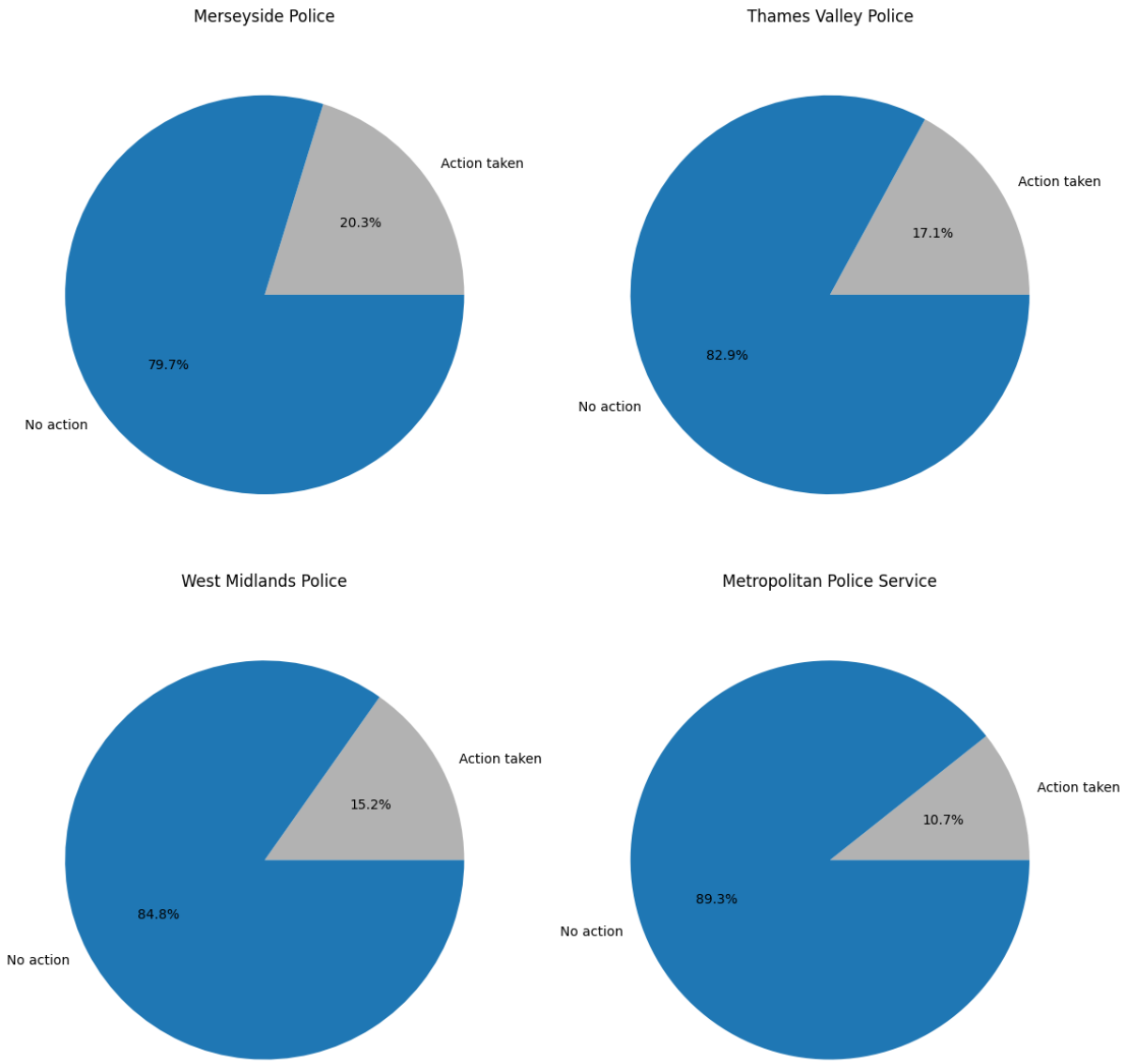
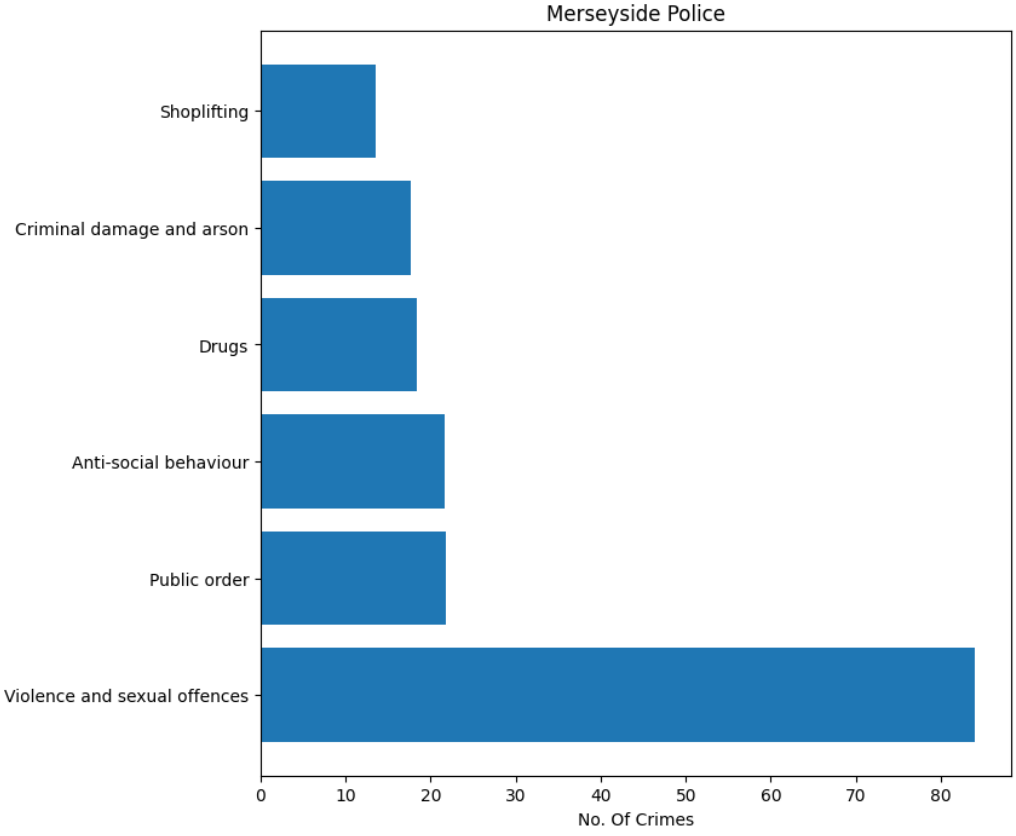
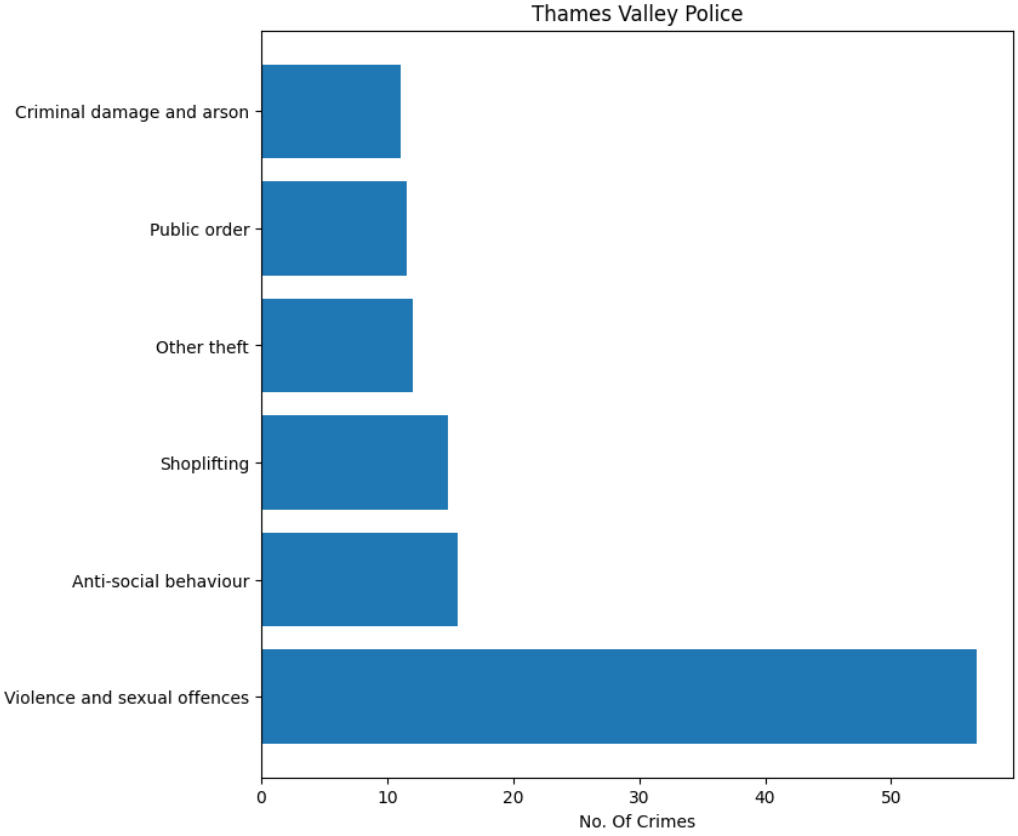


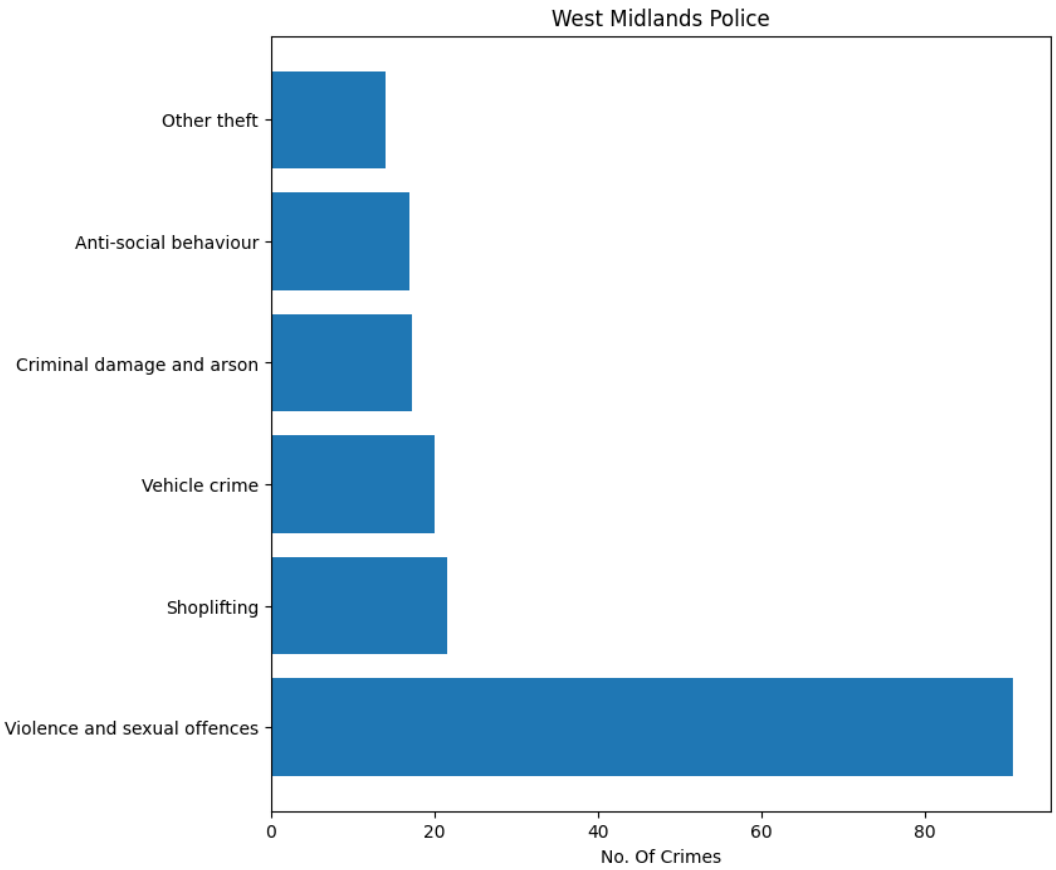
Figure 5: Comparing the proportion of action taken outcomes across all the police forces

It is interesting to note that out of all the chosen forces, the Metropolitan Police Service has a slightly larger number of No action outcomes (i.e. outcomes that don't result in a charge or arrest). This might be due to the fact that across the whole timeframe, a majority of action outcomes occur later in the timeframe and so over the whole timeframe the majority of outcomes had no action.

Top crimes across the forces







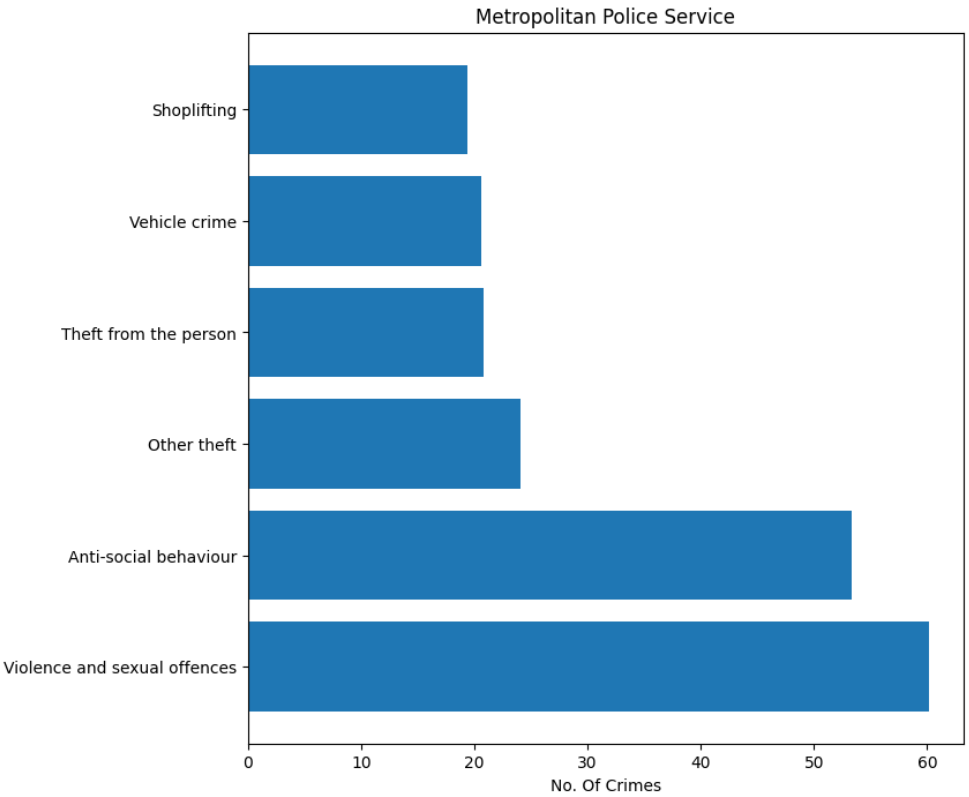
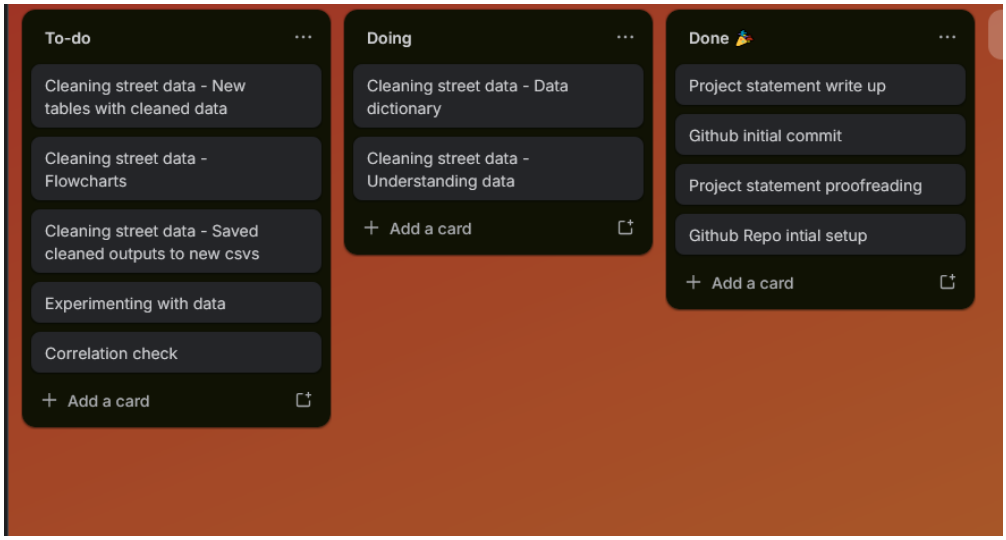


Figure 6: Comparing top 5 crime types across all police forces in the past 2 years

Across all the police forces studied, it is important to note that Violence and sexual offences ranked 1st, and with anti-social behaviour consistently being within the top 5. This might indicate that other types might be significantly under-reported or that there is a significant gap between crimes that are violent or sexual offences and any other type of offence committed. This kind of study would be further improved by considering each LSOA within each police force's jurisdiction.

Project management

The project had mainly met all key deadlines for each phase of the project except for the analysis phase which had taken an extra day to complete. This cascaded into preventing the final report writing from starting until the final day of work and hence resulted in a rushed completion of the project.



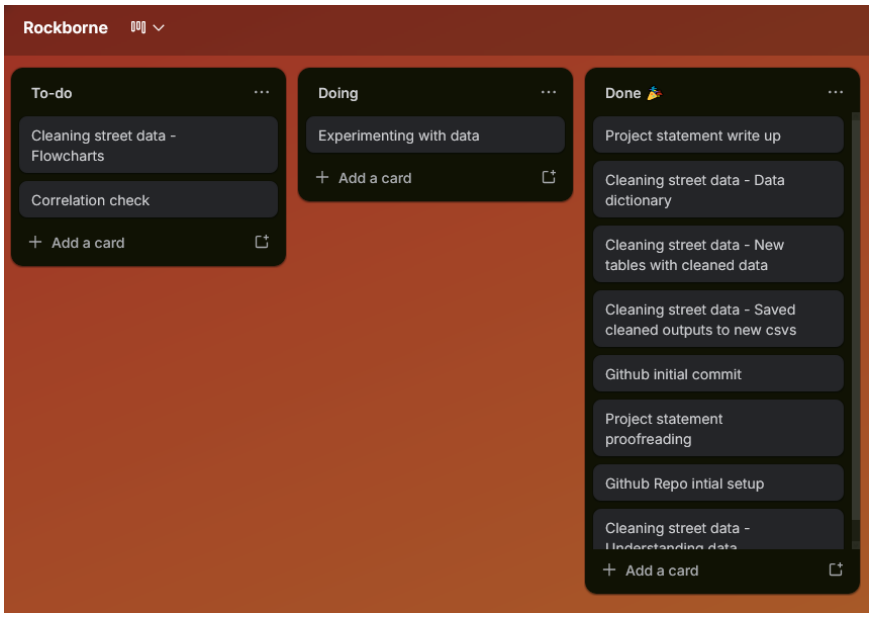


Figure 7: Progression of tasks across the project

Despite the rushed completion, the project met most of the tasks detailed in the original project statement. The problem from attempting to do a correlation analysis was the lack of continuous variables to do a correlation analysis between them as most of the data provided was categorical and hence not continuous.

Key takeaways and concluding remarks

As part of the preliminary data studies, it has been interesting to note that across the major city police forces (with the exception of Thames Valley), there is not much of a difference between the police forces reported per 1000 people and even in the top types of crimes being committed. The more important differences come from the changes in trend between the forces with the rapidly changing types of action taken versus the growth rate of crime across the recent 2-year period.

In relation to these findings the 2 services that are to be recommended for study are:

- Metropolitan Police Service: The rapid increase in action taken as well as the sheer scale of the area of greater London make it suitable for further study; the scale makes it suitable for comparisons within greater London on a borough-by-borough basis for better granularity.
- West Midlands Police: This force is similar to Metropolitan Police Service without the larger scale, the downwards trend of number of crimes recorded is of interest for further investigation.